

Python — Build Your First REST API with Flask

Lesson Notes & Core Concepts Guide

1. What is an API? Meet Waiter Flask-y!

An **API (Application Programming Interface)** acts as a messenger between different applications or systems. Think of ordering your favorite snack at a Dhaba:

1. The Client (YOU)

The customer placing the order for snacks and drinks.

2. The API (Waiter Flask-y)

Takes your request/order to the kitchen and brings your data back.

3. The Server (The Kitchen)

Where your data is prepared and stored.

2. Communication Format: JSON

JSON (JavaScript Object Notation) is the "secret menu card" that both clients and server computers understand effortlessly. It structures data in key-value pairs.

Key Concept: When a server sends data back to a client, it packages it in JSON format using Flask's built-in `jsonify()` utility function.

3. Step-by-Step Code Implementation

Below is the complete code to create a simple menu API endpoint in Flask:

```
from flask import Flask, jsonify

app = Flask(__name__)
```

```
@app.route('/menu')
def get_menu():
    indian_menu = {
        "snack": "Samosa",
        "drink": "Masala Chai",
        "sweet": "Gulab Jamun"
    }
    return jsonify(indian_menu)

if __name__ == '__main__':
    app.run(port=5000)
```

Testing Your API

Run your Python file and navigate to <http://localhost:5000/menu> in your web browser to view the JSON output returned by your route.

Student Practice & Review

Test your understanding of Flask REST APIs

Practice Questions

Answer the following questions based on the lesson notes. Write down your solutions clearly in your notebook.

Question 1 (Multiple Choice)

In the restaurant analogy, what does the API represent?

(A) The customer placing the order (B) The kitchen preparing the food (C) The waiter carrying orders and delivering food back (D) The dining table

Question 2 (Multiple Choice)

Which Flask helper function is used to convert a Python dictionary into a JSON response?

(A) `json.dumps()` (B) `jsonify()` (C) `convert_json()` (D) `render_template()`

Question 3 (Short Answer)

Explain why JSON is referred to as the "secret menu card" between the client and the server.

Question 4 (Coding Challenge)

Write a Flask route for `/my-team` that returns a JSON response containing the following key-value pairs:

• `team_name` • `captain` • `trophies_won` • `top_batsman`